

Challenge

Write a program to calculate the product of a number and its first 10 multiples.

Task

Given an integer n , print the first 10 multiples of n .
For example, if $n = 2$, the output should be:
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

Input Format

A single integer n .

Constraints

$2 \leq n \leq 20$

Output Format

Print 10 lines of output, each line i (where $1 \leq i \leq 10$) contains the result of $n \times i$ in the form:

$n \times i = \text{result}$

Sample Input

2

Sample Output

2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n;
5     scanf("%d",&n);
6     for(int i = 1; i <=10; i++)
7     {
8         printf("%d x %d = %d\n", n, i, n
9     }
10 }
```

	Input	Expected	Got	
✓	2	2 x 1 = 2 2 x 2 = 4 2 x 3 = 6 2 x 4 = 8 2 x 5 = 10 2 x 6 = 12 2 x 7 = 14 2 x 8 = 16 2 x 9 = 18 2 x 10 = 20	2 x 1 = 2 2 x 2 = 4 2 x 3 = 6 2 x 4 = 8 2 x 5 = 10 2 x 6 = 12 2 x 7 = 14 2 x 8 = 16 2 x 9 = 18 2 x 10 = 20	✓
Passed all tests! ✓				

Sample Input 1

Sample Output 1

Explanation 1

1. Cannot use item 1 because $k = 1$ and $sum \equiv k$ has to be avoided at any time.
2. Hence, max total is achieved by $sum = 0 + 2 = 2$.

Sample Case 2

Sample Input For Custom Testing

Sample Input 2

3

3

Sample Output 2

5

Explanation 2

$2 + 3 = 5$, is the best case for maximum nutrients.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,k,t=0;
5     scanf("%d%d",&n,&k);
6     for(int i=1;i<=n;i++){
7         t=(t+i)%1000000007;
8         if(t==k){
9             t -=1;
10        }
11    }
12    printf("%d",t);
13 }
```

	Input	Expected	Got	
✓	2 2	3	3	✓
✓	2 1	2	2	✓
✓	3 3	5	5	✓

Passed all tests! ✓

Sample Output 0

Explanation 0

Factoring $n = 10$ we get $\{1, 2, 5, 10\}$. We then return the $p = 5^{th}$ factor as our answer.

Sample Input 1

10
5

Sample Output 1

0

Explanation 1

Factoring $n = 10$ we get $\{1, 2, 5, 10\}$. There are only 4 factors and $p = 5$. We return 0 as our answer.

Sample Input 2

1
1

Sample Output 2

1

Explanation 2

Factoring $n = 1$ we get $\{1\}$. We then return the $p = 1^{st}$ factor as our answer.

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 int main()
3 {
4     int n,p,c=0;
5     scanf("%d%d",&n,&p);
6     for(int i=1;i<=n;i++){
7         if(n%i==0){
8             c++;
9         }
10        if(c==p){
11            printf("%d",i);
12            break;
13        }
14    }
15    if(c!=p){
16        printf("%d",0);
17    }
18    return 0;
19 }
```

	Input	Expected	Got	
✓	10 3	5	5	✓
✓	10 5	0	0	✓
✓	1 1	1	1	✓

Passed all tests! ✓